

RaceDay API Endpoint Plan

This document defines the REST API endpoints planned for the RaceDay system. The plan follows the finalized ERD, including Routes belonging to Categories, and will guide the Part 2 API implementation.

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new Participant account in the RaceDay system.	None	FullName, Email, Password, Phone	201 Created – Account created. 400 Bad Request – Invalid/missing information. 409 Conflict – Email exists.
POST	/api/auth/login	Authenticates a registered user and provides an authentication token.	None	Email, Password	200 OK – Login successful and token returned. 401 Unauthorized – Invalid credentials.

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Retrieves the profile of the currently logged-in user.	Any	None	200 OK – Profile returned. 401 Unauthorized – Authentication required.
PUT	/api/users/me	Updates the profile of the currently logged-in user.	Any	FullName, Email, Phone	200 OK – Profile updated. 400 Bad Request – Invalid information. 409 Conflict – Email exists.

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieves upcoming RaceDay events that users can browse.	None	None	200 OK – List of events returned.
GET	/api/events/{id}	Retrieves details of a specific RaceDay event.	None	None	200 OK – Event returned. 404 Not Found – Event does not exist.
POST	/api/events	Creates a new running, cycling or walking event.	Organiser	Name, Description, EventDate, Location, EventType	201 Created – Event created. 400 Bad Request – Invalid event information.
PUT	/api/events/{id}	Updates an event managed by the logged-in Organiser.	Organiser	Name, Description, EventDate, Location, EventType	200 OK – Event updated. 403 Forbidden – Not owner. 404 Not Found – Event does not exist.
DELETE	/api/events/{id}	Deletes an event managed by the logged-in Organiser.	Organiser	None	204 No Content – Event deleted. 403 Forbidden – Not owner. 404 Not Found – Event does not exist.

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Retrieves all categories belonging to an event.	None	None	200 OK – Categories returned. 404 Not Found – Event does not exist.
GET	/api/categories/{id}	Retrieves details of a specific category.	None	None	200 OK – Category returned. 404 Not Found – Category does not exist.
POST	/api/events/{eventId}/categories	Creates a category for an event.	Organiser	Name, DistanceKm, MaxParticipants, EntryFee	201 Created – Category created. 404 Not Found – Event does not exist.
PUT	/api/categories/{id}	Updates an existing category.	Organiser	Name, DistanceKm, MaxParticipants, EntryFee	200 OK – Category updated. 403 Forbidden – Not owner. 404 Not Found – Category does not exist.
DELETE	/api/categories/{id}	Deletes an event category.	Organiser	None	204 No Content – Category deleted. 403 Forbidden – Not owner. 404 Not Found – Category does not exist.

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/categories/{categoryId}/enrolments	Enrols the logged-in Participant and assigns a bib number.	Participant	None	201 Created – Enrolment created with assigned bib. 404 Not Found – Category missing. 409 Conflict – Already enrolled/category full.
GET	/api/users/me/enrolments	Retrieves the logged-in Participant's enrolments.	Participant	None	200 OK – Enrolments returned.
GET	/api/events/{eventId}/enrolments	Retrieves all participant enrolments for an event.	Organiser	None	200 OK – Enrolments returned. 404 Not Found – Event does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrolments/{id}	Retrieves details of a specific enrolment.	Any	None	200 OK – Enrolment returned. 403 Forbidden – Access denied. 404 Not Found – Enrolment missing.
PUT	/api/enrolments/{id}/status	Updates the status of an enrolment.	Organiser	Status	200 OK – Status updated. 404 Not Found – Enrolment missing.
DELETE	/api/enrolments/{id}	Allows a Participant to withdraw their own enrolment.	Participant	None	204 No Content – Enrolment cancelled. 403 Forbidden – Not their enrolment. 404 Not Found – Enrolment missing.

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/results	Records the official result for a participant enrolment.	Organiser	FinishTime, Position	201 Created – Result recorded. 404 Not Found – Enrolment missing. 409 Conflict – Result exists.
GET	/api/users/me/results	Retrieves the logged-in Participant's personal results.	Participant	None	200 OK – Personal results returned.
GET	/api/events/{eventId}/results	Retrieves results for an event.	Organiser	None	200 OK – Results returned. 404 Not Found – Event missing.
GET	/api/results/{id}	Retrieves a specific race result.	Any	None	200 OK – Result returned. 404 Not Found – Result missing.
PUT	/api/results/{id}	Updates or corrects an existing result.	Organiser	FinishTime, Position	200 OK – Result updated. 404 Not Found – Result missing.

7. Routes

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories/{categoryId}/routes	Retrieves route information belonging to a specific category.	None	None	200 OK – Route information returned. 404 Not Found – Category missing.
GET	/api/routes/{id}	Retrieves details of a specific route.	None	None	200 OK – Route returned. 404 Not Found – Route missing.
POST	/api/categories/{categoryId}/routes	Creates route information for a specific category.	Organiser	RouteName, DistanceKm, ElevationGain, MapUrl	201 Created – Route created. 404 Not Found – Category missing.
PUT	/api/routes/{id}	Updates an existing category route.	Organiser	RouteName, DistanceKm, ElevationGain, MapUrl	200 OK – Route updated. 403 Forbidden – Not owner. 404 Not Found – Route missing.
DELETE	/api/routes/{id}	Deletes route information from a category.	Organiser	None	204 No Content – Route deleted. 403 Forbidden – Not owner. 404 Not Found – Route missing.

Endpoint Summary: Authentication (2), User Profile (2), Events (5), Categories (5), Event Enrolments (6), Results (5), Routes (5) — **30 endpoints total.**